

Due: 8:30 pm, Mar 7, Friday

Upload a word file answering the questions and your original Python code to receive credit.

Find the measurements of rock samples posted on Canvas.

1. Remove all samples with missing measurements and report the number of removed samples.
2. Check for duplicate entries and remove them if they are errors. Report the number of samples removed.
3. Identify and remove samples with outlier porosity values. Show how the outliers were determined and report the number of samples removed.
4. Present the cross plots and histograms of all the cleaned properties.
5. Determine all the correlation coefficients.
6. Discuss whether the correlation coefficients capture the dependency in this dataset.
7. Use a regression method of your choice to determine porosity as a function of depth. Report R^2 and MSE.
8. Determine if porosity shows a stronger dependency on any other property than indicated in problem 7. If so, develop an alternative regression model and report its R^2 and MSE values.